

Make-Up

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!**

1. (20 pts) Determine the radius of convergence of each of the following series:

a.
$$\sum_{n=1}^{\infty} \frac{n^2 + 2n}{n+1} (3z+i)^{2n}$$

b.
$$\sum_{n=1}^{\infty} \left(1+2i-3i^{n^3-n}\right)^{n/2} (z-1+5i)^n$$

2. (20 pts) Let G be a region in $\mathbb{C}$ and let $f \in \mathcal{A}(G)$, $f = u + iv$. Prove that if $v = u^2$ on G , then f is constant on G .

3. (20 pts) Find all solutions of the equation $z^{2i} = -2 + 2i$.

4. (20 pts) Let $f(z) = e^{\frac{2z}{1-z}}$. Let T denote the unit circle, i.e., $T = \{z : |z| = 1\}$. Let γ be the subarc of T which goes, in the upper half-plane, from i to -1 . Identify and sketch the image of γ under f .

5. (10 pts) Let M be the Möbius transformation which maps $1, -i, i$ to $i, -i, 1$, resp. Find a formula for M .

6. (10 pts) The unit disk D , whose boundary is the circle C , which passes through the three points $i, 0, 1$, is divided into four congruent subregions by the lines

$y = 1 - x$ and $y = x$, say, D_1, D_2, D_3 and D_4 . (See figure to the right.)

Let M be the Möbius transformation

given by
$$M(z) = \frac{(1-i)z - i}{iz + 1 - i}.$$

Find/identify the images E_j of each subregion D_j under M , i.e., find

$E_j = M(D_j), \quad j = 1, 2, 3, 4.$

